

Mesh Architecture

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Introduction

As the number of processors in a parallel computer increases, providing efficient communication among processors becomes a significant challenge. A simple bus or ring network is insufficient because communication delays increase rapidly with the number of processors.

The **Mesh Architecture** addresses this problem by arranging processors in a regular two-dimensional (or multi-dimensional) grid. Each processor communicates directly with its neighboring processors through dedicated communication links.

The mesh topology is widely adopted because of its **simple structure, modular design, scalability, and ease of implementation**. It is one of the most commonly used static interconnection networks in parallel computers.

Definition

A **Mesh Architecture** is a **direct (static) interconnection network** in which processing elements are arranged in a two-dimensional or multi-dimensional grid, and each processor is connected to its immediate neighboring processors.

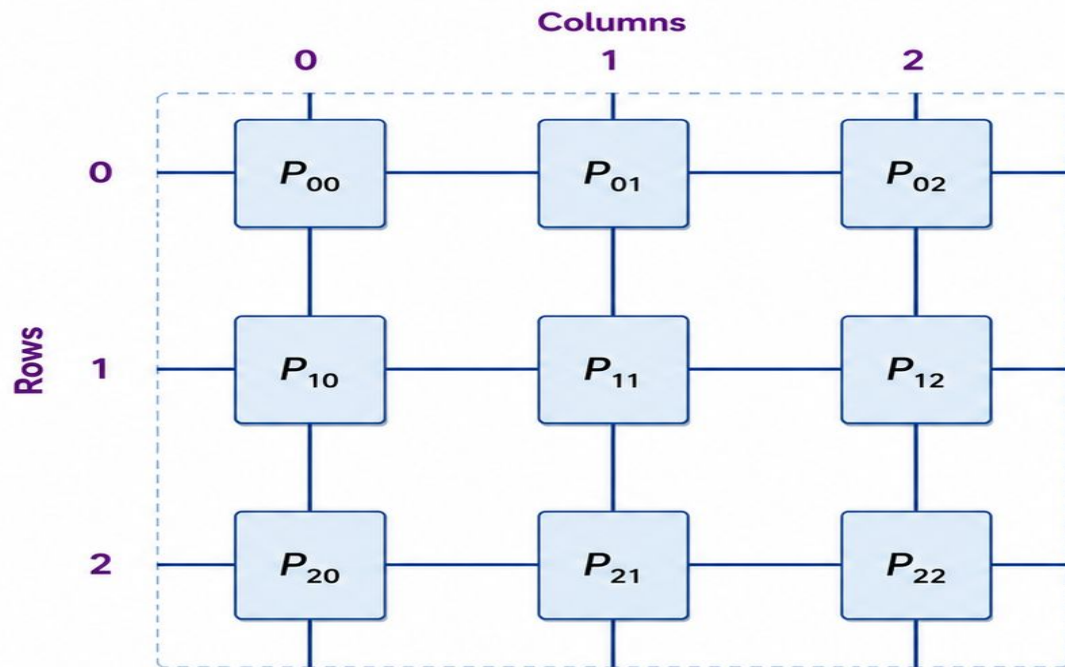
Each processor communicates only with its adjacent processors, while communication with distant processors occurs through intermediate nodes.

Characteristics

- Regular grid topology
- Fixed communication links
- Scalable architecture
- Supports parallel communication
- Suitable for large multiprocessor systems

Figure 2.1 – Basic Concept of Mesh Architecture

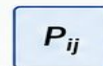
In a mesh architecture, processing elements (PEs) are arranged in a 2-D grid and each PE is connected to its immediate neighboring PEs.



Key Characteristics

- Processors are arranged in a 2-dimensional grid.
- Each processor is connected to its immediate neighbors (up, down, left, right).
- Corner nodes have 2 neighbors.
- Edge nodes have 3 neighbors.
- Interior nodes have 4 neighbors.

LEGEND



Processing Element (PE)



Communication Link



Mesh architecture provides a simple, regular and scalable interconnection structure suitable for building large parallel and distributed systems.

Figure 2.2 – Why Mesh Architecture?

Mesh architecture overcomes the limitations of simple interconnection networks and provides efficient, scalable and parallel communication.

1



Scalability

Processors can be added row-wise or column-wise without redesigning the entire system.

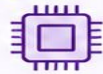
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Reduced Communication Bottleneck

Multiple processor pairs can communicate simultaneously without interfering with each other.

3



Simple Hardware

Each processor requires only a few communication links (up to 4 in 2-D mesh).

4



Parallel Data Transfer

Different data can be transferred concurrently along different paths.

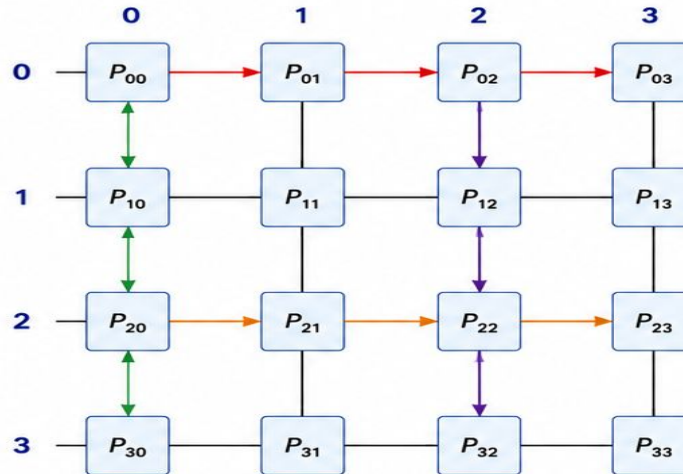
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Modular Expansion

Additional processors can be incorporated by extending the mesh.

Example: 4 × 4 Mesh



Multiple communications can occur simultaneously

- Communication between P_{00} and P_{03}
- Communication between P_{00} and P_{30}
- Communication between P_{02} and P_{32}
- Communication between P_{20} and P_{23}

Why Mesh is Better?

- Enables high parallelism and concurrency.
- Provides multiple paths for data transfer.
- Improves system throughput.
- Offers good balance between performance and hardware cost.
- Suitable for large-scale parallel and distributed systems.

Typical Use Cases

- ✓ Multicomputers
- ✓ Network-on-Chip (NoC)
- ✓ GPU architectures
- ✓ AI accelerators
- ✓ Supercomputers



In Summary: Mesh architecture provides a scalable, efficient, and cost-effective solution that supports concurrent communications and is ideal for large parallel processing systems.

3. Organization of Mesh Architecture

In a mesh architecture, processing elements (PEs) are arranged in a two-dimensional grid (rows $\times$ columns). Each PE is connected to its immediate neighboring PEs through communication links.

Key Points

- Let the number of rows be m and the number of columns be n .
- Total number of processing elements (PEs):
$$N = m \times n$$
- Each PE is identified by its coordinates (row, column) starting from (0, 0) at the top-left corner.
- Connections are only between neighboring PEs (Up, Down, Left, Right).

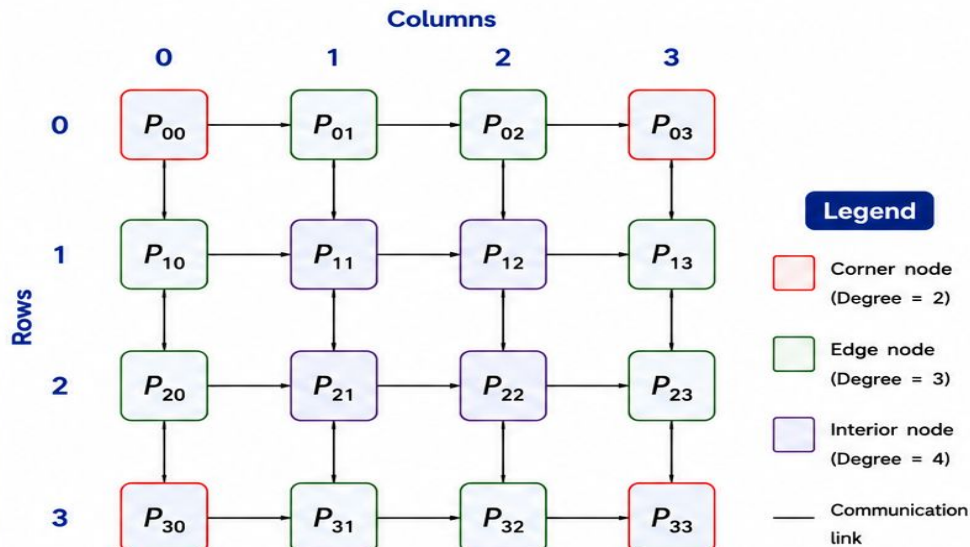
Types of Nodes

- Interior node** : Has 4 neighbors (Up, Down, Left, Right).
- Edge node** : Has 3 neighbors.
- Corner node** : Has 2 neighbors.
- In a 2-D mesh, the degree of a node can be 2, 3 or 4.

Example

- Consider a mesh with $m = 4$ rows and $n = 4$ columns.
- Total PEs, $N = 4 \times 4 = 16$.
- Each PE is connected to its adjacent PEs as shown in Figure 2.3.

Figure 2.3 4 $\times$ 4 Mesh Organization



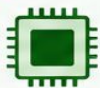
Properties of 4 $\times$ 4 Mesh

- Total number of PEs (N) = $4 \times 4 = 16$
- Interior nodes = $(m-2) \times (n-2) = (4-2) \times (4-2) = 4$
- Edge nodes (excluding corners) = $2(m+n-4) = 2(4+4-4) = 8$
- Corner nodes = 4
- Degree of nodes = 2, 3 or 4
- Maximum possible neighbors (any interior node) = 4

4. Components of Mesh Architecture

A mesh architecture consists of several key components that work together to enable efficient computation and communication among processing elements.

1 Processing Elements (PEs)



- Each PE performs computation and processing.
- Arranged in a 2-D grid (rows × columns).
- Has a unique address (row, column).

2 Communication Links



- Bidirectional links connect each PE to its neighbors (Up, Down, Left, Right).
- Carry data, control and synchronization signals.
- Each link has a certain bandwidth and latency.

3 Router (Switch) at Each Node



- Routes incoming data packets to the appropriate output link.
- Implements routing algorithm (e.g., XY routing).
- Handles arbitration, flow control and buffering.

4 Local Memory



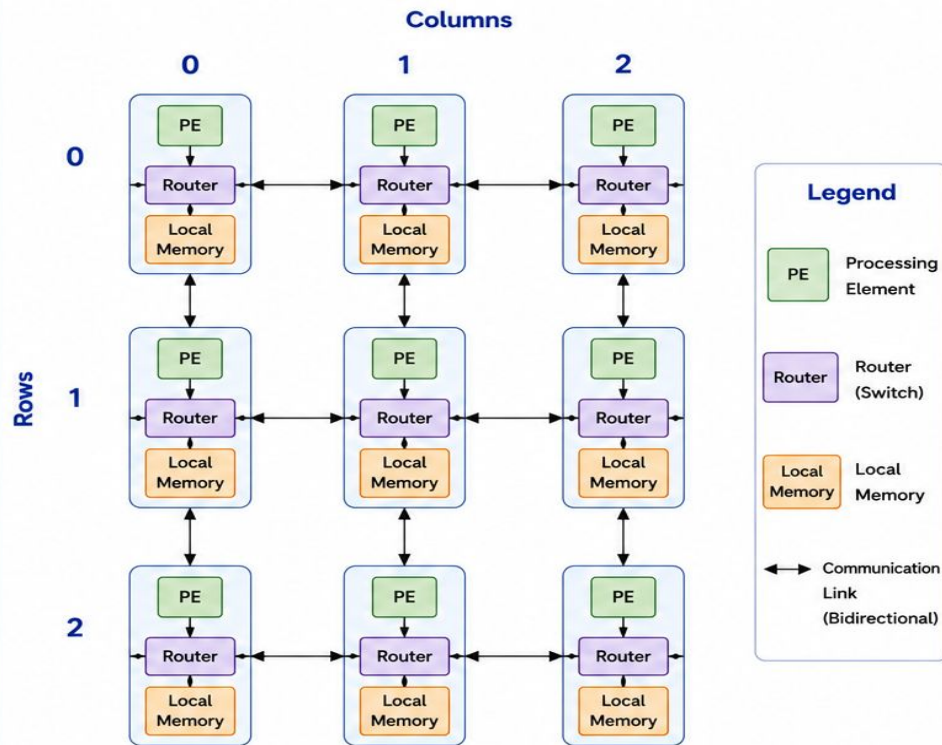
- Each PE has its own local memory.
- Stores instructions, input data and intermediate results.
- Reduces the need for global memory access and improves performance.

Summary



The combination of Processing Elements, Communication Links, Routers and Local Memory forms the basic building blocks of a mesh architecture, enabling scalable and efficient parallel processing.

Figure 2.4 Components of Mesh Architecture



Key Points

- Each node contains a Processing Element (PE), a Router (Switch) and Local Memory.
- Routers connect the node to its neighbors through communication links.
- Communication is possible in four directions: Up, Down, Left and Right.
- This modular structure enables high scalability and parallel communication.

5. Addressing Scheme

In a mesh architecture, each processing element (PE) is uniquely identified by its position in the grid using (Row, Column) coordinates. This addressing scheme simplifies routing and communication.

Key Points

- Each PE in an $m \times n$ mesh is identified by an ordered pair (i, j) .
- Here, i is the row number and j is the column number.
- Rows are numbered from 0 to $m - 1$ (top to bottom).
- Columns are numbered from 0 to $n - 1$ (left to right).
- The top-left processor has address $(0, 0)$.
- The bottom-right processor has address $(m - 1, n - 1)$.
- This scheme is simple, regular and easy to implement in hardware.

Addressing Rules

- Total number of processors (PEs) = $m \times n$
- Row range : $0 \leq i \leq m - 1$
- Column range : $0 \leq j \leq n - 1$
- Address of any processor = (i, j)
- Two processors are neighbors if their row and column differ by 1 in one direction.

Example

For a 4×4 mesh:

- Top-left processor = $(0, 0)$
- Processor in 2nd row, 1st column (0-based) = $(1, 0)$
- Processor in 3rd row, 2nd column = $(2, 1)$
- Bottom-right processor = $(3, 3)$

General Address
(Row, Column)

Benefits

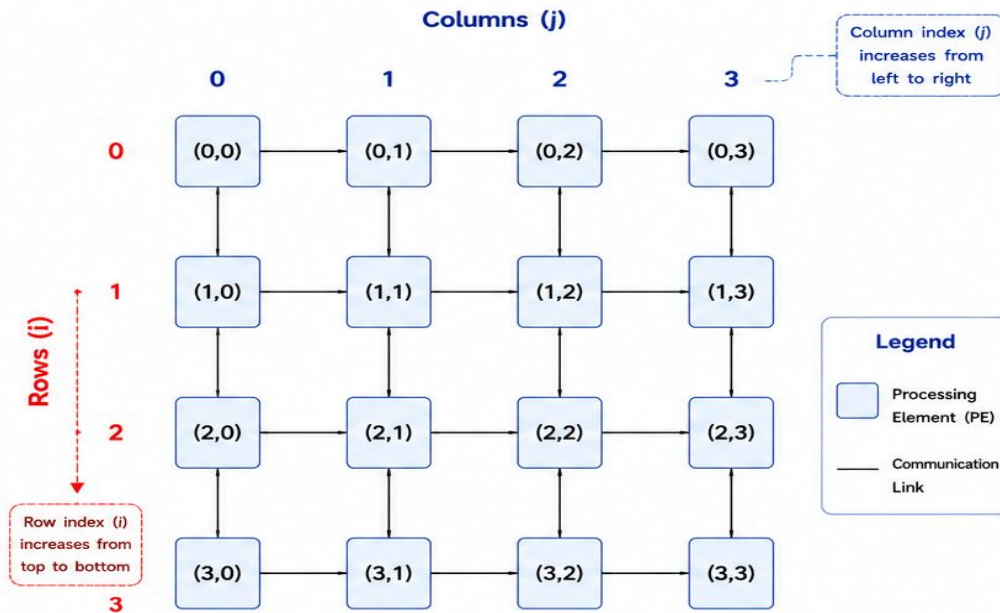


- Simplifies routing algorithms (e.g., XY Routing).
- Easy mapping of processors to coordinates.
- Scalable and regular addressing for large meshes.



Conclusion: Using (Row, Column) coordinates provides a simple, regular and efficient addressing scheme which forms the basis for routing and communication in mesh architectures.

Figure 2.5 Processor Addressing in a Mesh



Example

- Address of top-left processor = $(0, 0)$
- Address of processor in 2nd row, 1st column = $(1, 0)$
- Address of processor in 3rd row, 2nd column = $(2, 1)$
- Address of bottom-right processor = $(3, 3)$



Note

In an $m \times n$ mesh, the address of any processor is (i, j) where $0 \leq i \leq m - 1$ and $0 \leq j \leq n - 1$

6. Communication in Mesh Networks

In a mesh network, a processor communicates with another processor by sending data through intermediate neighboring processors (hops) until the destination is reached.

Key Points

- Each processor can directly communicate only with its immediate neighbors (Up, Down, Left, Right).
- If the destination is not a neighbor, the data is forwarded through a sequence of intermediate processors.
- Communication is performed hop-by-hop.
- The path taken depends on the routing algorithm used (e.g., XY Routing).
- Multiple paths may exist between two processors; routing decides the actual path.

How Communication Works

- Source processor generates data for the destination.
- Data is sent to a neighbor based on the routing algorithm.
- That neighbor forwards the data to the next node.
- This continues until the data reaches the destination processor.
- An acknowledgment (ACK) may be sent back if required.

Example

- Consider a 4×4 mesh.
- Source (S) = $P(0,0)$
- Destination (D) = $P(2,3)$
- One possible communication path using XY Routing (move in X direction first, then Y direction):

$P(0,0) \rightarrow P(1,0) \rightarrow P(2,0) \rightarrow P(2,1) \rightarrow P(2,2) \rightarrow P(2,3)$

Important Notes

- The number of hops determines the communication delay.
- Longer paths result in higher latency.
- Efficient routing minimizes the number of hops and avoids congestion.

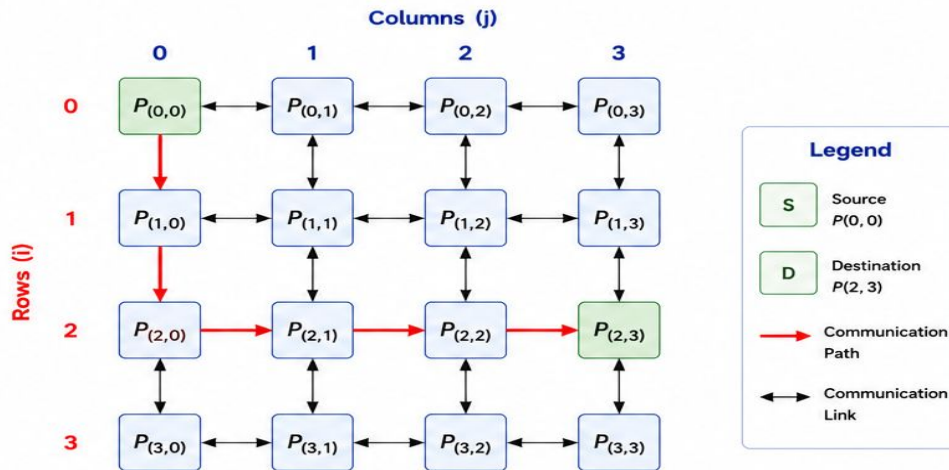
Summary



Communication in mesh networks is achieved by forwarding data through neighboring nodes in multiple hops according to a routing algorithm until the destination is reached.

Figure 2.6 Communication Path in a Mesh Network (4×4 Mesh)

Example: Communication from Source $P(0,0)$ to Destination $P(2,3)$ using XY Routing



Path Details

Path: $P(0,0) \rightarrow P(1,0) \rightarrow P(2,0) \rightarrow P(2,1) \rightarrow P(2,2) \rightarrow P(2,3)$
Number of Hops = 5
Total Links Traversed = 5
Movement = First in X direction (Down), then in Y direction (Right)

Other Possible Paths (Examples)

- Path 1 (XY Routing - as shown above)
 - Path 2: $P(0,0) \rightarrow P(0,1) \rightarrow P(0,2) \rightarrow P(0,3) \rightarrow P(1,3) \rightarrow P(2,3)$ (6 hops)
 - Path 3: $P(0,0) \rightarrow P(1,0) \rightarrow P(1,1) \rightarrow P(1,2) \rightarrow P(1,3) \rightarrow P(2,3)$ (6 hops)
- Note: XY Routing gives the minimum number of hops in this case.



Efficient routing reduces the number of hops, minimizes delay, and improves overall network performance.

7. Routing Algorithms (XY Routing)

Routing determines the path that data takes from a source processor to a destination processor. The most widely used routing algorithm in mesh networks is the XY Routing (also called Dimension-Order Routing).

XY Routing – Basic Idea

In a 2-D mesh, communication is performed in two phases:

- 1 X-Direction Phase (Row Direction):**
Move first along the rows (up or down) until the source and destination are in the same row.
- 2 Y-Direction Phase (Column Direction):**
Then move along the columns (left or right) until the destination is reached.

Properties of XY Routing

- Deterministic – the path is uniquely determined by source and destination.
- Deadlock-free – packets never wait in a circular dependency.
- Simple to implement – requires only local information.
- Minimal number of turns – at most one (from X phase to Y phase).

Example

- Consider a 4×4 mesh.
- Source $S = P(0, 0)$
- Destination $D = P(2, 3)$
- Using XY Routing:

- 1** Move in X-direction (rows): from row 0 down to row 2.
- 2** Move in Y-direction (columns): from column 0 right to column 3.

Advantages



- Simple and easy to implement.
- Ensures deadlock-free communication.
- Provides shortest path in terms of number of hops for 2-D meshes.
- Widely used in NoC and parallel computer architectures.

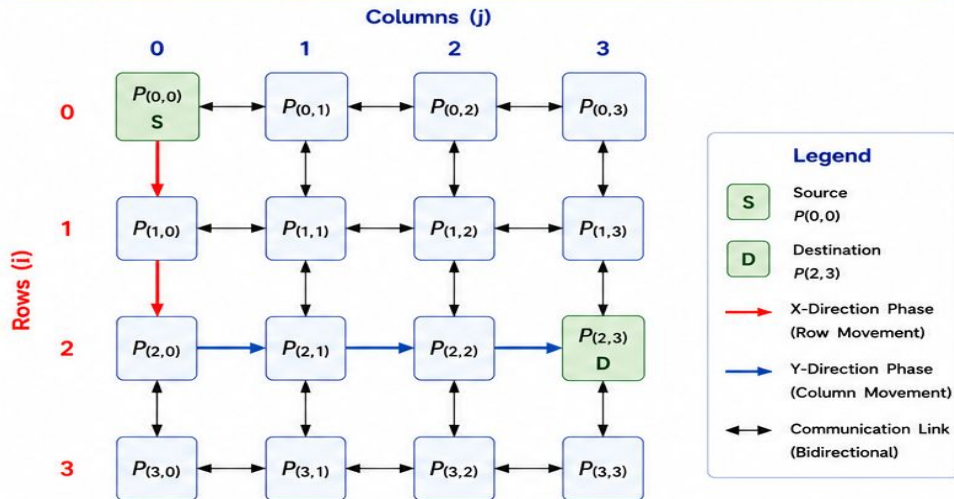
Note



For an $m \times n$ mesh, XY Routing first reduces the difference in row index (i), then reduces the difference in column index (j).

Figure 2.7 XY Routing in a 4×4 Mesh

Example: Source $P(0,0)$ to Destination $P(2,3)$ using XY Routing



Routing Steps

- 1** Start at Source $P(0,0)$.
- 2** **X-Direction Phase** (move in rows): $P(0,0) \rightarrow P(1,0) \rightarrow P(2,0)$.
- 3** **Y-Direction Phase** (move in columns): $P(2,0) \rightarrow P(2,1) \rightarrow P(2,2) \rightarrow P(2,3)$.
- 4** Destination $P(2,3)$ is reached.

Path and Performance

- Path followed = $P(0,0) \rightarrow P(1,0) \rightarrow P(2,0) \rightarrow P(2,1) \rightarrow P(2,2) \rightarrow P(2,3)$
- Number of Hops = 5
- Total Links Traversed = 5
- Number of Turns = 1 (at $P(2,0)$, from vertical to horizontal)
- This is the minimum number of hops in a 4×4 mesh for this pair.

8. Performance Analysis of Mesh Architecture

The performance of a mesh interconnection network is evaluated using several metrics that measure its size, connectivity and communication efficiency.

Key Performance Metrics

1 Number of Nodes (Processors)

$$N = m \times n$$

Total number of processing elements in an $m \times n$ mesh.

2 Node Degree

- Interior node = 4
- Edge node = 3
- Corner node = 2

Number of direct neighbors of a node.

3 Network Diameter

$$D = (m - 1) + (n - 1)$$

Maximum shortest distance between any two nodes.

4 Average Distance

$$\bar{d} \approx \frac{m+n}{2}$$

Average number of hops between two nodes.

5 Bisection Width

$$B = \min(m, n)$$

Minimum number of links that must be cut to split the network into two equal halves.

6 Total Number of Links

$$L = 2mn - m - n$$

Total bidirectional communication links in the mesh.

Example: 4 × 4 Mesh

- Rows (m) = 4
- Columns (n) = 4
- Number of nodes (N) = $4 \times 4 = 16$
- Network diameter (D) = $(4 - 1) + (4 - 1) = 6$

- Average distance = $\frac{(4+4)}{2} = 4$
- Bisection width = $\min(4, 4) = 4$
- Total links = $2(4 \times 4) - 4 - 4 = 24$

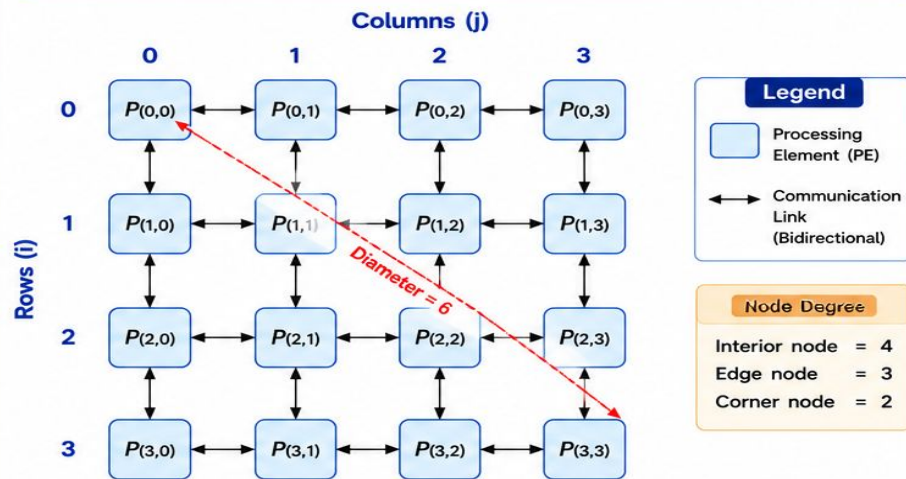
Key Takeaways



- Mesh architecture offers a good balance between cost and performance.
- The diameter grows linearly with the network size.
- Bisection width increases with the number of rows and columns, enabling more parallel communication.
- Regular structure provides predictable and uniform performance.

Figure 2.8 Performance Metrics of a Mesh Network

Performance metrics illustrated for a 4 × 4 mesh architecture



Network Diameter

$$D = (m - 1) + (n - 1)$$

For 4 × 4 mesh:

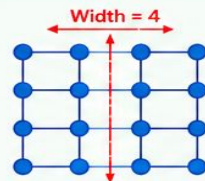
$$D = (4 - 1) + (4 - 1) = 6$$

Bisection Width

$$B = \min(m, n)$$

For 4 × 4 mesh:

$$B = \min(4, 4) = 4$$



Average Distance

$$\bar{d} \approx \frac{m+n}{2}$$

For 4 × 4 mesh:

$$\bar{d} = \frac{4+4}{2} = 4$$

Total Number of Links

$$L = 2mn - m - n$$

For 4 × 4 mesh:

$$L = 2(4 \times 4) - 4 - 4 = 24$$

Number of Nodes

$$N = m \times n$$

For 4 × 4 mesh:

$$N = 4 \times 4 = 16$$

9. Advantages of Mesh Architecture

Mesh architecture offers several advantages that make it suitable for large-scale parallel and distributed systems.

Key Advantages

- 1 **High Scalability** – Easily supports a large number of processors by adding more rows and columns.
- 2 **Regular Structure** – Simple, uniform and regular topology makes design and implementation easy.
- 3 **Fault Tolerance** – Multiple paths exist between any two nodes, so the network can continue to operate even if some links fail.
- 4 **High Bandwidth** – Many links operate in parallel, providing high aggregate bandwidth.
- 5 **Low Latency (Short Paths)** – XY routing ensures short and predictable paths with minimal number of hops.
- 6 **Deadlock-Free Routing** – Dimension-order (XY) routing is inherently deadlock-free.
- 7 **Cost-Effective for Large Systems** – Uses simple point-to-point links instead of expensive global switches.
- 8 **Good Balance of Cost and Performance** – Provides an excellent trade-off between performance, scalability and hardware cost.

10. Limitations of Mesh Architecture

Despite its benefits, mesh architecture also has some limitations.

Key Limitations

- 1 **Increased Path Length with Size** – As the network grows, the average distance (number of hops) between nodes increases.
- 2 **Higher Number of Links** – Requires a large number of point-to-point links, which increases hardware complexity and cost.
- 3 **Scalability Overhead** – The number of links grows linearly with the number of nodes, leading to more wiring and area.
- 4 **Congestion** – Hot spots may occur when many nodes communicate through the same intermediate nodes.
- 5 **Power Consumption** – More links and routers increase the overall power consumption.
- 6 **Physical Layout Challenges** – In very large meshes, wire length and delay between distant nodes can become significant.

Note



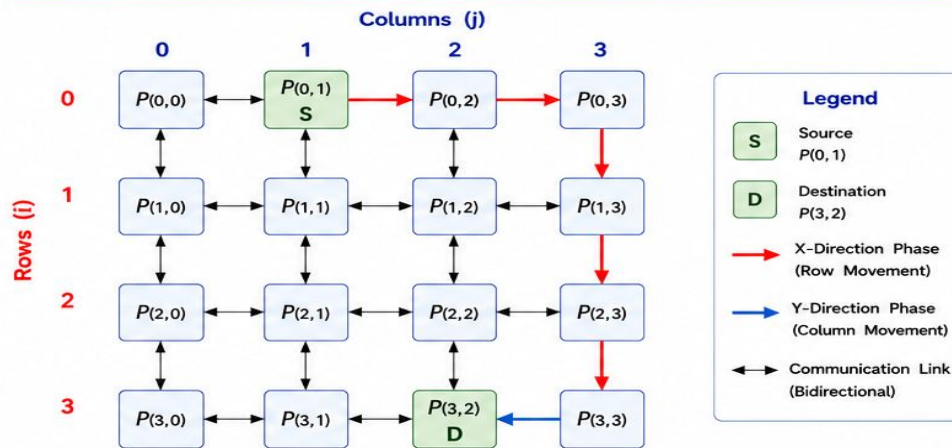
The effectiveness of a mesh architecture depends on the application communication pattern, routing algorithm and the hardware implementation of routers and links.



Conclusion: Mesh architecture provides a regular, scalable and efficient communication network. With proper routing (XY Routing), it ensures deadlock-free communication and good performance, making it a preferred choice for many large-scale systems.

Figure 2.9 Example: 4 × 4 Mesh with XY Routing Path

Example: Communication from source $P(0, 1)$ to destination $P(3, 2)$ using XY Routing.



Path followed (XY Routing)

X-Direction (Rows) : $P(0, 1) \rightarrow P(0, 2) \rightarrow P(0, 3) \rightarrow P(1, 3) \rightarrow P(2, 3) \rightarrow P(3, 3)$

Y-Direction (Columns) : $P(3, 3) \rightarrow P(3, 2)$

Total Hops = 6 (5 moves in X-direction + 1 move in Y-direction)

11. Applications of Mesh Architecture

Mesh architecture is widely used in many modern computing systems because of its scalability and cost-effectiveness.



Multi-core Processors



Network-on-Chip (NoC)



Parallel Computers



Data Centers and Clusters



AI / ML Accelerators



High Performance Computing (HPC)